

IROBMAN Project Report: $SE(3)$ Error-State EKF for 6D Object Pose Tracking

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Abstract—Accurate and robust 6D object pose tracking is essential for robotic manipulation in dynamic environments. FoundationPose provides accurate object pose estimation and tracking through iterative pose refinement. However, when the target undergoes fast motion, a poor pose initialization may require additional refinement iterations, increasing the computational cost, while insufficient refinement can lead to tracking failure. FoundationPose++ addresses this problem by introducing a Kalman-filter-based motion prediction mechanism that fuses pose observations with motion predictions to provide a better initialization for FoundationPose at each frame. In addition, a 2D tracker is incorporated to correct the predicted object location in the image plane, improving robustness to rapid translational motion and reducing the dependence on costly pose refinement.

However, the Kalman filtering framework in FoundationPose++ represents 3D orientation using Euler angles, which does not fully respect the geometric structure of rigid-body motion and may introduce representation-related issues during rotational motion. In this project, the original filtering framework is reformulated as an error-state Extended Kalman Filter (ES-EKF) on $SE(3)$. The nominal object pose is maintained directly on $SE(3)$, while local pose uncertainty and Kalman corrections are represented in the corresponding Lie algebra. This formulation enables rotation and translation to be handled within a unified rigid-body motion framework without directly filtering Euler angles. The original 2D tracking and measurement-fusion strategy of FoundationPose++ is retained and integrated consistently into the proposed error-state framework. The resulting system combines real-time RGB-D input, 2D object localization, $SE(3)$ -based temporal filtering, and FoundationPose pose refinement to provide robust initialization for 6D object tracking, particularly under rapid object motion.

Index Terms—6D object pose tracking, FoundationPose++, $SE(3)$, error-state Extended Kalman Filter, Lie group, RGB-D.

I. INTRODUCTION

Reliable 6D object pose tracking is an important component of many robotic applications, including manipulation, grasping, and human–robot interaction. In dynamic environments, a tracking system is required not only to estimate the three-dimensional position and orientation of an object accurately, but also to maintain stable tracking when the object undergoes rapid motion. This remains challenging for pose tracking methods that rely strongly on the pose estimate from the previous frame.

FoundationPose provides a general framework for 6D object pose estimation and tracking using RGB-D observations and object geometry. During continuous tracking, the pose from the previous frame is used as an initialization and is iteratively refined according to the current observation. This strategy

works well when the inter-frame motion is relatively small. However, for rapidly moving objects, the previous pose can differ significantly from the current pose. The resulting poor initialization makes pose refinement more difficult and may require additional refinement iterations, increasing the computational cost. If sufficient refinement cannot be performed within the available processing time, the tracker may fail to recover the object pose and eventually lose the target.

FoundationPose++ improves the robustness of this tracking process by introducing Kalman filtering and additional 2D object tracking. The Kalman filter combines previous pose observations with prediction results obtained from the motion model to update the object pose, thereby providing FoundationPose with a more suitable initialization than simply reusing the pose from the previous frame. In addition, a 2D tracker, such as Cutie or OTrack, provides object localization in the image plane. This additional measurement is fused into the filtering process to correct the predicted image-plane position. Together, motion prediction and 2D localization improve the initialization supplied to FoundationPose and reduce the dependence on a large number of refinement iterations, which is particularly useful under rapid object motion.

Despite these improvements, the filtering formulation used in FoundationPose++ represents the rigid-body pose using Cartesian translation and Euler angles and applies a conventional linear Kalman filter to these variables. Although convenient, this representation treats the pose as a vector in a Euclidean space and does not explicitly account for the underlying geometry of three-dimensional rigid-body motion. In particular, rotations belong to the Lie group $SO(3)$, while a complete rigid-body pose naturally belongs to the special Euclidean group $SE(3)$. This motivates a filtering formulation in which pose prediction and correction are consistent with this geometric structure.

In this project, the Kalman filtering component of FoundationPose++ is reformulated as an error-state Extended Kalman Filter (ES-EKF) on $SE(3)$. Instead of directly filtering Euler-angle pose variables, the nominal rigid-body pose is maintained on $SE(3)$, while local pose errors and their uncertainty are represented in the corresponding Lie algebra. The filter therefore estimates a local error state, which is used to correct the nominal pose through the exponential map. The original 2D tracking mechanism of FoundationPose++ is retained, while its measurement update is reformulated consistently within the new error-state framework.

The main contributions of this project are therefore three-fold:

- 1) Reformulating the Euler-angle-based linear Kalman filtering module of FoundationPose++ as an error-state EKF on $SE(3)$;
- 2) Integrating the original 2D tracker measurement into the proposed Lie-group-based filtering framework;
- 3) Evaluating the proposed formulation in simulation, offline RGB-D tracking, and real-time ZED experiments, including dedicated rotational stress tests around the $\pm 180^\circ$ Euler branch boundary and the near- 90° singular configuration.

II. FOUNDATIONPOSE++

FoundationPose++ combines FoundationPose with temporal motion prediction and 2D object tracking to improve pose initialization during continuous 6D object tracking. FoundationPose itself represents the object pose as a rigid transformation, whereas the original Kalman filter represents translation and orientation using Euclidean coordinates.

For the first frame, FoundationPose performs pose registration using the initial object mask. The resulting pose initializes the Kalman filter. For each subsequent frame, the predicted state is corrected using the available FoundationPose pose estimate and the current 2D tracking result. The corrected pose is then used as the initialization for FoundationPose refinement, after which the filter predicts the state for the next iteration.

A. Euler-Angle Kalman Filter

The original FoundationPose++ Kalman filter uses the 12-dimensional state

$$\mathbf{x}_k = [\mathbf{p}_k \quad \boldsymbol{\theta}_k \quad \mathbf{v}_k \quad \dot{\boldsymbol{\theta}}_k]^T, \quad (1)$$

where $\mathbf{p}_k$ is translation, $\boldsymbol{\theta}_k = [r_x, r_y, r_z]^T$ contains the XYZ Euler coordinates, and $\mathbf{v}_k$ and $\dot{\boldsymbol{\theta}}_k$ are the corresponding translational and angular rates. In particular, r_y denotes the *middle Euler coordinate* (pitch).

The rotation matrix obtained from FoundationPose is converted to XYZ Euler coordinates before entering the filter. The 6D pose measurement is therefore

$$\mathbf{z}_k = \begin{bmatrix} \mathbf{t}_{\text{meas}} \\ \boldsymbol{\theta}_{\text{meas}} \end{bmatrix}, \quad (2)$$

and the innovation is computed directly in Euclidean coordinates:

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{H}\mathbf{x}_k^-. \quad (3)$$

The standard Kalman correction is then applied to the state and covariance. Consequently, rotational disagreement is represented by direct subtraction of XYZ Euler coordinates.

FoundationPose++ additionally uses a 2D tracker to localize the object in the image. The tracked bounding-box center is

back-projected using the available depth and camera intrinsics, providing an additional measurement of the x - and y -translation components. This measurement is fused through a second Kalman correction.

After the 6D and 2D updates, the corrected translation and Euler coordinates are converted back into a rigid transformation and supplied to FoundationPose as the initialization for current-frame pose refinement.

For motion prediction, the original filter assumes constant translational and Euler-coordinate rates:

$$\mathbf{x}_{k+1}^- = \begin{bmatrix} \mathbf{I}_6 & \Delta t \mathbf{I}_6 \\ \mathbf{0}_6 & \mathbf{I}_6 \end{bmatrix} \mathbf{x}_k^+. \quad (4)$$

Thus, rotational motion is predicted by linear extrapolation of Euler coordinates rather than by composition of rotations.

B. Motivation for an $SE(3)$ Formulation

The Euler-angle representation is convenient, but it does not provide a globally well-conditioned coordinate system for three-dimensional rotation. XYZ Euler coordinates contain branch discontinuities and a singular configuration when the middle Euler coordinate r_y approaches $\pm 90^\circ$. Therefore, direct subtraction or extrapolation of these coordinates can produce large coordinate changes that do not correspond to equally large physical rotations.

A rigid-body pose naturally belongs to the Lie group $SE(3)$. This motivates maintaining the nominal pose directly as a rigid transformation and expressing only local pose errors in Euclidean coordinates.

III. $SE(3)$ ERROR-STATE EKF

The original temporal filter is reformulated as an error-state Extended Kalman Filter (ES-EKF) on $SE(3)$. The overall FoundationPose++ processing order is retained:

6D pose update $\rightarrow$ 2D tracker update $\rightarrow$
FoundationPose refinement $\rightarrow$ motion prediction.

The difference lies in how pose error, correction, and prediction are represented.

A. Nominal Pose and Local Error State

The nominal object pose is represented directly by

$$\mathbf{T}_k = \begin{bmatrix} \mathbf{R}_k & \mathbf{p}_k \\ \mathbf{0}^T & 1 \end{bmatrix} \in SE(3). \quad (5)$$

Instead of treating the complete pose as a Euclidean vector, a local pose perturbation is represented in the Lie algebra by

$$\delta \boldsymbol{\xi}_k = \begin{bmatrix} \delta \boldsymbol{\rho}_k \\ \delta \boldsymbol{\phi}_k \end{bmatrix} \in \mathbb{R}^6, \quad (6)$$

where $\delta \boldsymbol{\rho}_k$ and $\delta \boldsymbol{\phi}_k$ denote local translational and rotational perturbations, respectively.

Together with linear- and angular-velocity errors, the 12-dimensional error state is

$$\delta \mathbf{x}_k = \begin{bmatrix} \delta \boldsymbol{\xi}_k \\ \delta \mathbf{v}_k \\ \delta \boldsymbol{\omega}_k \end{bmatrix}. \quad (7)$$

A left-invariant error convention is used. The true pose is related to the nominal pose by

$$\mathbf{T}_{k,\text{true}} = \text{Exp}_{SE(3)}(\delta \boldsymbol{\xi}_k) \mathbf{T}_k. \quad (8)$$

Thus, the local pose error acts on the nominal pose through left multiplication.

B. Pose Measurement and Correction

Let $\mathbf{T}_k^-$ be the predicted pose and $\mathbf{T}_{\text{obs}}$ the pose observation provided by FoundationPose. Rather than converting both rotations to Euler coordinates and subtracting them, the pose innovation is computed from their relative transformation:

$$\mathbf{r}_k = \text{Log}_{SE(3)}(\mathbf{T}_{\text{obs}}(\mathbf{T}_k^-)^{-1}). \quad (9)$$

This residual describes a local rigid-body discrepancy in the Lie algebra. The standard Kalman update is performed on the local error state, after which the estimated pose correction is injected into the nominal pose as

$$\mathbf{T}_k^+ = \text{Exp}_{SE(3)}(\delta \hat{\boldsymbol{\xi}}_k) \mathbf{T}_k^-. \quad (10)$$

The translational and angular velocity components are corrected additively, and the error state is reset around the corrected nominal state.

The original 2D tracking measurement is retained. The tracked image-plane location is back-projected using depth and camera intrinsics, and the resulting x - y position discrepancy is used as an additional measurement update. Its Jacobian is defined with respect to the local left-invariant error state, so the original 2D tracking functionality is preserved within the $SE(3)$ formulation.

After these updates, the corrected nominal pose $\mathbf{T}_k^+$ is supplied directly to FoundationPose as the initialization for current-frame pose refinement. The ES-EKF therefore does not replace FoundationPose; it replaces the temporal pose-filtering and prediction representation used to initialize FoundationPose.

C. $SE(3)$ Motion Prediction

After refinement, the nominal pose is propagated using a constant spatial velocity model. Defining

$$\mathbf{u}_k = \begin{bmatrix} \mathbf{v}_k \\ \boldsymbol{\omega}_k \end{bmatrix}, \quad (11)$$

the pose prediction is

$$\mathbf{T}_{k+1}^- = \text{Exp}_{SE(3)}(\mathbf{u}_k \Delta t_k) \mathbf{T}_k^+. \quad (12)$$

The spatial linear and angular velocities are assumed constant during one prediction interval. In the real-time implementation, Δt_k is obtained from the measured interval between consecutive frames.

The covariance is propagated in the corresponding local error coordinates using the linearized error-state dynamics. Unlike the Euler-angle filter, however, the nominal pose itself is never propagated by linearly extrapolating global orientation coordinates.

The essential distinction between the two formulations is therefore that the Euler-angle KF performs rotational prediction and correction directly on XYZ Euler coordinates, whereas the $SE(3)$ ES-EKF maintains the nominal pose on the rigid-body group and represents rotational disagreement as a local Lie-algebra error.

IV. EXPERIMENTS AND RESULTS

To evaluate the proposed $SE(3)$ error-state extended Kalman filter (ES-EKF) under different tracking conditions, three experiments are conducted. The first experiment investigates object pose tracking during robotic manipulation, where a dexterous hand moves a cube in simulation. Accurate ground-truth poses are directly available from the simulation environment, enabling a quantitative evaluation of the tracking accuracy. The second experiment evaluates the offline tracking performance on an RGB-D image sequence to examine the behavior of the proposed method under different motion and observation conditions. The third experiment evaluates the filtering formulations under real-world RGB-D tracking conditions using a ZED camera. In addition to verifying real-time operation of the proposed $SE(3)$ ES-EKF, dedicated rotational stress tests are conducted around the $\pm 180^\circ$ Euler branch boundary and the near- 90° singular configuration of the middle Euler coordinate r_y . These experiments are used to investigate representation-related behaviour that is not exposed by the nominal-motion experiments.

A. Experiment I: Robot Manipulation Tracking in Simulation

The first experiment evaluates the object pose tracking accuracy in a simulated robotic manipulation scenario. A dexterous robotic hand is used to manipulate a cube in the MuJoCo simulation environment. During the manipulation, observations are recorded using a fixed virtual RGB-D camera, resulting in a sequence of 504 frames. The RGB observations are stored as an AVI video, while the corresponding depth measurements are stored as frame-wise depth images. During the motion, the cube is partially occluded by the dexterous hand for short periods. Therefore, the experiment considers both continuous object motion and pose tracking under mild partial occlusion.

Since the complete state of the simulated environment is directly accessible in MuJoCo, the ground-truth pose of the cube with respect to the camera can be obtained for every frame. These poses provide an accurate reference trajectory for quantitatively evaluating the estimated 6D object poses.

The same recorded RGB-D data are then processed using two tracking configurations. The first configuration uses the original Euler-angle-based Kalman filter in FoundationPose++, while the second configuration uses the proposed $SE(3)$ ES-EKF. In both configurations, Cutie is used as the 2D

TABLE I
TRANSLATION ERRORS IN THE ROBOT MANIPULATION EXPERIMENT.

Method	MAE (mm)	RMSE (mm)	Max (mm)
Euler-angle KF	0.390	0.423	0.869
$SE(3)$ ES-EKF	0.389	0.422	0.879

TABLE II
ROTATION ERRORS IN THE ROBOT MANIPULATION EXPERIMENT.

Method	MAE (deg)	RMSE (deg)	Max (deg)
Euler-angle KF	0.702	0.734	1.566
$SE(3)$ ES-EKF	0.703	0.735	1.561

object tracker, followed by the same FoundationPose pose refinement procedure. Except for the filtering method, the input data, tracking components, and associated parameters are kept identical to ensure a fair comparison between the two filtering approaches.

Representative tracking results from the simulated manipulation sequence are shown in Fig. 1. The estimated object pose is visualized by projecting the object coordinate axes and 3D bounding box onto the RGB image.

The tracking accuracy is evaluated separately for translation and rotation. For frame k , the translation error is defined as

$$e_t(k) = \|\mathbf{t}_k^{\text{est}} - \mathbf{t}_k^{\text{gt}}\|_2, \quad (13)$$

where $\mathbf{t}_k^{\text{est}}$ and $\mathbf{t}_k^{\text{gt}}$ denote the estimated and ground-truth object translations, respectively.

The rotational error is evaluated using the geodesic distance on $SO(3)$,

$$e_R(k) = \cos^{-1} \left(\frac{\text{tr}[(\mathbf{R}_k^{\text{gt}})^T \mathbf{R}_k^{\text{est}}] - 1}{2} \right), \quad (14)$$

where $\mathbf{R}_k^{\text{est}}$ and $\mathbf{R}_k^{\text{gt}}$ are the estimated and ground-truth rotation matrices. The resulting angular error is converted from radians to degrees for reporting. The quantitative translation and rotation results over the complete sequence are summarized in Tables I and II.

The Average Distance of Model Points (ADD) metric was additionally used to evaluate the overall 6D pose accuracy. For each frame, ADD was computed as

$$e_{\text{ADD}} = \frac{1}{M} \sum_{i=1}^M \left\| (\hat{\mathbf{R}}\mathbf{x}_i + \hat{\mathbf{t}}) - (\mathbf{R}^{\text{gt}}\mathbf{x}_i + \mathbf{t}^{\text{gt}}) \right\|_2, \quad (15)$$

where $\mathbf{x}_i$ denotes a point on the object model, M is the number of sampled model points, $(\hat{\mathbf{R}}, \hat{\mathbf{t}})$ is the estimated pose, and $(\mathbf{R}^{\text{gt}}, \mathbf{t}^{\text{gt}})$ is the ground-truth pose. To obtain a representative sampling of the object geometry, $M = 5000$ points were uniformly sampled from the surface of the object mesh. The same sampled point set was used for both filtering methods.

TABLE III
ADD ERRORS IN THE ROBOT MANIPULATION EXPERIMENT.

Method	Mean (mm)	RMSE (mm)	Max (mm)
Euler-angle KF	0.570	0.594	1.021
$SE(3)$ ES-EKF	0.569	0.593	1.023

The quantitative results in Tables I, II, and III show that the two filtering approaches achieve nearly identical tracking accuracy. The Euler-angle KF obtains a translation MAE of 0.390 mm, a rotation MAE of 0.702°, and a mean ADD of 0.570 mm, while the proposed $SE(3)$ ES-EKF obtains 0.389 mm, 0.703°, and 0.569 mm, respectively. The differences between the two methods are therefore negligible under the motion conditions considered in this experiment.

This result indicates that no evident degradation associated with Euler-angle discontinuities or singular configurations is observed during the relatively smooth and continuous motion of this sequence. Consequently, the Euler-angle KF remains effective under these nominal conditions. The motivation for the proposed $SE(3)$ formulation is therefore not to guarantee lower tracking errors in every sequence, but to provide a geometrically consistent treatment of rigid-body pose prediction, error representation, and correction.

B. Experiment II: RGB-D Image-Sequence Tracking

The second experiment evaluates the tracking performance on an offline RGB-D sequence containing relatively rapid object motion. The input sequence consists of frame-wise RGB images and their corresponding depth images. A textured 3D mesh of the target LEGO block is provided to FoundationPose, together with an initial segmentation mask in the first frame for pose initialization. The complete sequence contains 116 frames.

Compared with the relatively continuous object motion in Experiment I, this sequence contains faster motion and larger inter-frame pose changes, providing a more challenging tracking condition. In particular, larger inter-frame motion can increase the discrepancy between the predicted object pose and the current RGB-D observation.

The same sequence is processed using the two filtering configurations considered in Experiment I. The first configuration uses the original Euler-angle-based Kalman filter, while the second uses the proposed $SE(3)$ ES-EKF. Cutie is used as the 2D object tracker in both configurations, while the remaining components and parameters of the tracking pipeline are kept unchanged. Since ground-truth poses are not available for this sequence, the experiment focuses on a qualitative comparison of the tracking behavior rather than a quantitative evaluation of pose accuracy.

Representative tracking results are shown in Fig. 2. Frames 20, 40, 60, 80, and 100 are selected to illustrate the tracking behavior at different stages of the sequence.

As illustrated in Fig. 2, both filtering configurations successfully track the LEGO block throughout the selected frames. No evident qualitative difference in tracking behavior is observed



Fig. 1. Representative pose tracking results in the MuJoCo robotic manipulation experiment. The top row shows the Euler-angle KF results, while the bottom row shows the proposed $SE(3)$ ES-EKF results. The selected frames illustrate pose tracking during continuous manipulation and partial occlusion by the dexterous hand.

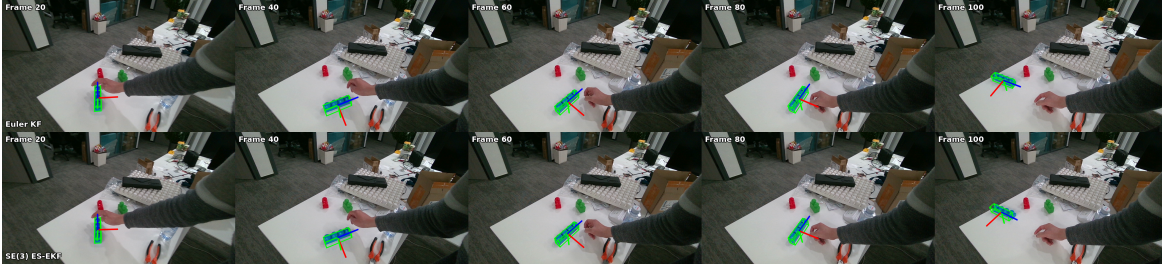


Fig. 2. Qualitative comparison of pose tracking on the offline RGB-D sequence. The top row shows the results obtained using the Euler-angle KF, while the bottom row shows the results obtained using the proposed $SE(3)$ ES-EKF. Both configurations maintain continuous tracking throughout the selected frames despite the relatively rapid object motion.

between the Euler-angle KF and the proposed $SE(3)$ ES-EKF under the tested motion conditions.

These results demonstrate that the proposed $SE(3)$ ES-EKF can be integrated into the FoundationPose++ tracking pipeline while maintaining stable tracking behavior in this RGB-D sequence. However, because ground-truth poses are not available, this experiment does not provide a quantitative comparison of pose accuracy. The results should therefore be interpreted as a qualitative evaluation showing that both filtering configurations maintain continuous tracking, rather than as evidence that either filtering method achieves higher pose accuracy.

C. Experiment III: Real-Time Tracking and Rotational Stress Tests

The third experiment evaluates the two filtering formulations under real-world RGB-D tracking conditions using a ZED camera. In addition to demonstrating real-time operation, this experiment is designed to examine two distinct representation-related difficulties of XYZ Euler angles: the $\pm 180^\circ$ branch boundary and the near- 90° singular configuration of the middle Euler coordinate r_y (pitch). These are different phenomena. The former is caused by the periodic branch convention of Euler coordinates, whereas the latter is a parameterization singularity in which the two outer Euler coordinates become strongly coupled.

The real-world recordings are not externally motion-captured and the Euler and $SE(3)$ stress-test sequences are independent rather than frame-synchronized executions of

identical motion. Consequently, the results below are used to characterize the observed behaviour of the two formulations rather than as a paired real-world pose-accuracy benchmark.

1) *Real-Time Tracking*: The proposed $SE(3)$ ES-EKF was integrated with FoundationPose and OTrack and evaluated using a ZED RGB-D camera. Two objects, a mustard bottle and a banana, were used to assess the sustained online operation of the complete tracking pipeline under ordinary manual object motion.

Three approximately two-minute real-time tracking trials were recorded for each object, giving six trials in total. Across all trials, 21,109 frames were processed.

The runtime results were highly consistent across the two objects. Across all 21,109 frames, the complete pipeline achieved a mean end-to-end rate of 29.35 FPS. FoundationPose and OTrack accounted for most of the per-frame computation, requiring 47.09% and 27.57% of the mean frame time, respectively.

In comparison, the $SE(3)$ EKF update required only 0.534 ms per frame and the $SE(3)$ motion prediction required 0.440 ms. Together, the proposed filtering operations required 0.973 ms per frame, corresponding to only 2.86% of the total mean frame time. These results show that the $SE(3)$ error-state formulation introduces only a small computational overhead relative to FoundationPose and OTrack while maintaining approximately 29.3 FPS in the complete real-time tracking pipeline.

2) $\pm 180^\circ$ Euler Branch-Crossing Test: A first rotational stress test examines continuous unidirectional rotation through

TABLE IV
RUNTIME SUMMARY FOR REPEATED REAL-TIME $SE(3)$ ES-EKF TRACKING.

Runtime statistic	Mustard bottle	Banana	Overall
Trials	3	3	6
Frames	10,539	10,570	21,109
Mean frame time	34.13 ms	34.03 ms	34.08 ms
Mean FPS	29.30	29.39	29.35
FoundationPose	16.071 ms (47.09%)	16.023 ms (47.09%)	16.047 ms (47.09%)
2D tracker (OSTrack)	9.399 ms (27.54%)	9.393 ms (27.60%)	9.396 ms (27.57%)
$SE(3)$ EKF update	0.534 ms (1.56%)	0.533 ms (1.57%)	0.534 ms (1.57%)
$SE(3)$ prediction	0.439 ms (1.29%)	0.440 ms (1.29%)	0.440 ms (1.29%)
$SE(3)$ filtering total	0.973 ms (2.85%)	0.974 ms (2.86%)	0.973 ms (2.86%)

TABLE V
REPEATED MULTI-TURN EULER BRANCH-CROSSING TRIALS.

Turns	Crossings	Max $ r_z $	Median $\Delta\theta$	Max crossing $\Delta\theta$
2	2	361.19°	1.39°	34.71°
3	3	349.72°	2.01°	42.63°
4	4	360.30°	1.98°	30.80°

TABLE VI
DYNAMIC NEAR-SINGULARITY STRESS-TEST SUMMARY.

Formulation	Outcome	Max innovation	Max output $\Delta\theta$
Euler-angle KF	Lost in 4/4	263.56–449.90°	74.68–106.73°
$SE(3)$ ES-EKF	Maintained in 2/2	16.63–18.33°	15.63–16.04°

the Euler branch boundary. Although the physical orientation changes continuously, an Euler coordinate represented on a fixed branch can switch from approximately $+180^\circ$ to -180° , or vice versa. Direct subtraction of the two coordinate values can therefore generate a residual close to 360° even though the actual local rotation is small.

Figure 3 illustrates a representative two-turn Euler-angle-KF recording. During continuous physical rotation, the Euler coordinate crosses its branch boundary and changes between values near $+180^\circ$ and -180° . Because the original filter forms its rotational innovation by directly subtracting Euler coordinates, this coordinate discontinuity can appear internally as a residual close to 360° .

Repeated trials containing two, three, and four continuous turns were analysed. An Euler branch-crossing event was detected when the absolute r_z innovation exceeded 180° . For a crossing detected at frame k , the associated output-pose transient was characterized by

$$\Delta\theta_{\text{cross}} = \max(\Delta\theta_{k \rightarrow k+1}, \Delta\theta_{k+1 \rightarrow k+2}), \quad (16)$$

where each $\Delta\theta$ is the frame-to-frame geodesic change on $SO(3)$.

Across the three Euler trials, nine branch-crossing events were detected. The maximum $|r_z|$ innovations remained close to 360° , while the median frame-to-frame output-pose changes were only about 1.4° – 2.0° . Pronounced output-pose transients were observed around the detected crossings, with maximum crossing-associated changes between 30.80° and 42.63° .

Independent $SE(3)$ recordings did not exhibit an analogous near- 360° coordinate residual. Their maximum local rotational innovations in the corresponding multi-turn stress tests were 11.28° , 14.40° , and 13.31° for the two-, three-, and four-turn recordings, respectively. Because these are independent recordings, the magnitudes characterize the behaviour of each

representation and are not frame-synchronized accuracy measurements.

3) *Near-90° Euler Singularity*: A second real-world stress test examines the XYZ Euler singularity that occurs as the middle Euler coordinate r_y (pitch) approaches $\pm 90^\circ$. This is distinct from the $\pm 180^\circ$ branch boundary. In the XYZ Euler parameterization, $\cos(r_y)$ approaches zero near the singular configuration. The two outer coordinates r_x and r_z then become strongly coupled and are no longer independently well-conditioned. Consequently, a small and smooth physical orientation change can correspond to large, coupled changes in the Euler-coordinate representation.

Two complementary tests are used. First, dynamic recordings move the object through the near-singular region. Four Euler-angle-KF recordings subsequently lost tracking, whereas two independent $SE(3)$ ES-EKF stress-test recordings maintained tracking. The observed ranges are summarized in Table VI.

The closest measured $|r_y|$ values in the four Euler trials were between 88.18° and 89.73° . Large Euler-coordinate innovations and large output-orientation transients were observed in every trial. The largest transient was not always temporally coincident with the closest measured approach to 90° ; therefore, these recordings are interpreted as near-singularity stress tests rather than evidence that crossing one particular numerical threshold deterministically causes tracking loss.

4) *Controlled Near-Singularity Diagnostic*: To separate representation effects from rapid physical motion, an additional recording holds the object approximately stationary with the measured middle Euler coordinate r_y (pitch) close to the singular configuration. A control recording holds the object at an orientation farther from the singularity.

Near the singular configuration, the median measured value is $r_y = 87.74^\circ$. Although the object is approximately stationary, the converted Euler coordinates exhibit

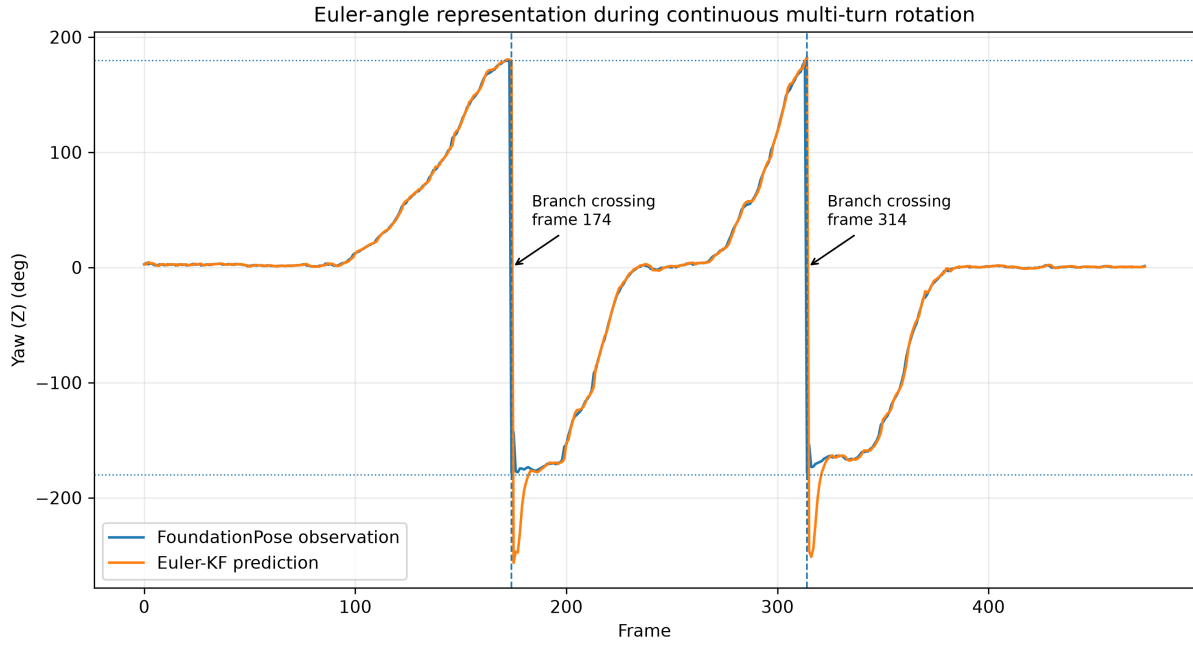


Fig. 3. Representative two-turn Euler branch-crossing experiment. Although the physical object rotation is continuous, the Euler-coordinate representation crosses the $\pm 180^\circ$ branch boundary. Direct subtraction of the Euler coordinates therefore produces near- 360° rotational innovations at the detected crossings, together with corresponding transient changes in the filter output pose.

$$\sigma(r_x) = 22.46^\circ, \quad \sigma(r_z) = 22.33^\circ, \quad (17)$$

with

$$\rho(r_x, r_z) = 0.9999. \quad (18)$$

For the control orientation, whose median middle coordinate is $r_y = -28.57^\circ$, the corresponding standard deviations are only 0.57° and 0.49° . These values describe variation of the Euler coordinates; they do not imply that the physical object itself oscillates by approximately 22° . Instead, the diagnostic exposes the poor conditioning of the XYZ Euler parameterization near its singular configuration.

Figure 4 visualizes this effect directly. When the measured middle Euler coordinate r_y is close to 90° , r_x and r_z vary almost along a one-dimensional coupled direction, despite the object being held approximately stationary. In contrast, at the control orientation farther from the singular configuration, both coordinates remain concentrated within a much smaller range. The figure therefore illustrates a coordinate-conditioning effect rather than a physical 22° oscillation of the object.

The coordinate effect also propagates through the Euler-angle KF because the original filter predicts and corrects these Euler coordinates directly. Table VII summarizes the measured internal filter quantities.

In particular, the standard deviations of the predicted r_x and r_z coordinates increase to approximately 31° , and the corresponding angular-rate states reach approximately 10.7° per frame. Because these rate states are reused by the next prediction step, the representation sensitivity is not confined

TABLE VII
PROPAGATION OF NEAR-SINGULAR EULER-COORDINATE VARIATION THROUGH THE EULER-ANGLE KF.

Quantity	Near 90°	Control
Measurement $\sigma(r_x)/\sigma(r_z)$	22.46/22.33 $^\circ$	0.57/0.49 $^\circ$
Predicted $\sigma(r_x)/\sigma(r_z)$	31.48/31.33 $^\circ$	0.69/0.61 $^\circ$
Median rotational innovation	24.44 $^\circ$	0.87 $^\circ$
95th-percentile innovation	97.84 $^\circ$	1.95 $^\circ$
Corrected $\sigma(r_x)/\sigma(r_z)$	21.68/21.56 $^\circ$	0.55/0.46 $^\circ$
Angular-rate $\sigma(\dot{r}_x)/\sigma(\dot{r}_z)$	10.74/10.69 $^\circ$ /frame	0.19/0.19 $^\circ$ /frame

to a single converted FoundationPose measurement; it can propagate through the innovation, correction, motion state, and subsequent prediction.

A component-wise comparison of r_x , r_y , and r_z is not appropriate for the $SE(3)$ ES-EKF because its rotational error state is not an XYZ Euler-angle state. The common quantity of interest is instead the local rotational discrepancy between predicted and observed orientations,

$$\phi_k = \text{Log}_{SO(3)}(\mathbf{R}_{\text{obs}} \mathbf{R}_{\text{pred}}^T). \quad (19)$$

The independent static near- 90° recordings are summarized in Table VIII.

The Euler and $SE(3)$ recordings in this comparison are independent sequences, and their innovations follow the different residual definitions of the two filters. The values are therefore used to characterize how each formulation behaves in the same near-singularity regime rather than as frame-by-frame pose errors.

Euler-Coordinate Behaviour Near the XYZ Euler Singularity

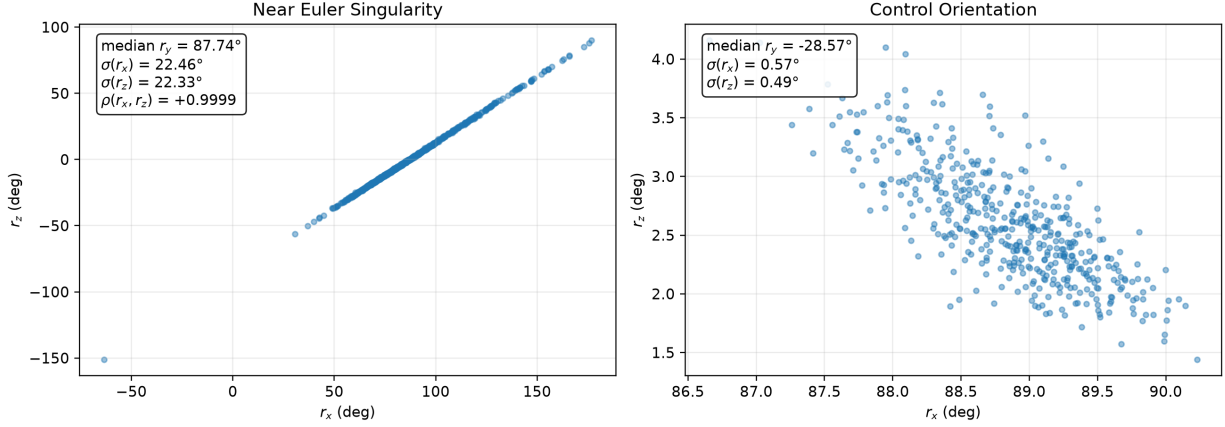


Fig. 4. Euler-coordinate behaviour in the approximately stationary near-90° diagnostic. Left: near the XYZ Euler singular configuration, with median middle Euler coordinate $r_y = 87.74^\circ$, the outer coordinates exhibit large and strongly coupled variation, $\sigma(r_x) = 22.46^\circ$, $\sigma(r_z) = 22.33^\circ$, and $\rho(r_x, r_z) = 0.9999$. Right: at the control orientation ($r_y = -28.57^\circ$), the corresponding variations are only 0.57° and 0.49° . These quantities describe Euler-coordinate variation and do not imply equivalent physical rotation of the stationary object.

TABLE VIII
STATIC NEAR-90° DIAGNOSTIC.

Quantity	Euler-angle KF	$SE(3)$ ES-EKF
Median rotational innovation	24.44°	0.89°
95th percentile	97.84°	2.07°
Maximum	266.87°	3.05°
Median output-pose $SO(3)$ change	1.14°	0.72°
Maximum output-pose $SO(3)$ change	5.49°	2.52°

D. Discussion

The experiments distinguish nominal tracking accuracy from representation robustness. In Experiment I, where accurate simulation ground truth is available and the motion remains smooth, the Euler-angle KF and the proposed $SE(3)$ ES-EKF achieve nearly identical translation, rotation, and ADD errors. This shows that the proposed formulation should not be interpreted as a method that universally reduces pose error.

Experiment II provides an offline RGB-D evaluation under faster motion without ground-truth poses. Both formulations can be integrated into the same FoundationPose++ pipeline and are evaluated qualitatively under the recorded sequence. Because no external ground truth is available, conclusions from this experiment are restricted to observed tracking behaviour rather than absolute pose accuracy.

The repeated real-time trials in Experiment III demonstrate that the proposed formulation can be incorporated into the complete tracking pipeline without becoming its computational bottleneck. Across six trials and 21,109 frames, the system achieved a mean rate of 29.35 FPS. The $SE(3)$ filtering operations required only 0.973 ms per frame, corresponding to 2.86% of the total mean frame time, while most of the

computation remained in FoundationPose and OTrack.

The rotational stress tests expose differences specifically associated with the orientation representation. At the $\pm 180^\circ$ branch boundary, direct Euler-coordinate subtraction can create a near-360° residual even though the physical orientation remains continuous. Near the 90° singularity of the middle Euler coordinate r_y (pitch), the two outer Euler coordinates become strongly coupled. The approximately stationary diagnostic shows that this behaviour is not simply a consequence of fast physical motion: large, highly correlated r_x and r_z variation is observed even while the object is held approximately stationary, and this variation propagates into the Euler-KF prediction, innovation, corrected state, and angular-rate state.

The $SE(3)$ ES-EKF avoids direct filtering of global Euler coordinates. Its rotational disagreement is represented locally in the Lie algebra through the relative rotation between the predicted and observed orientations. The main benefit demonstrated by the experiments is therefore geometric consistency and representation robustness, rather than a universal reduction in tracking error.

Several limitations remain. Accurate ground-truth poses are available only in simulation. The offline LEGO and real-world ZED experiments therefore rely primarily on qualitative tracking behaviour and internal filter diagnostics. The real-world stress tests also contain a limited number of objects and independently recorded trials; they do not constitute a frame-synchronized accuracy benchmark. Moreover, tracking failure in the complete vision pipeline can also depend on image quality, occlusion, object motion, depth measurements, and FoundationPose estimation errors. Future work should include externally measured real-world ground truth, synchronized

repeated trajectories, and a broader range of objects and motion conditions.

V. CONCLUSION

This project reformulated the Euler-angle-based temporal filtering component of FoundationPose++ as an error-state Extended Kalman Filter on $SE(3)$. The nominal rigid-body pose is maintained directly on the Lie group, while local pose errors and Kalman corrections are represented in the corresponding Lie algebra. The original 2D measurement-fusion strategy is retained, and the resulting corrected pose is used as the initialization for FoundationPose refinement.

Under smooth motion with simulation ground truth, the Euler-angle KF and the proposed $SE(3)$ ES-EKF achieve nearly identical pose accuracy. The experiments therefore do not support a claim that the $SE(3)$ formulation universally reduces tracking error. Instead, clearer differences appear when the orientation approaches problematic regions of the XYZ Euler representation.

At the $\pm 180^\circ$ branch boundary, the Euler filter can produce near- 360° coordinate residuals despite continuous physical rotation. Near the 90° singularity of the middle Euler coordinate r_y (pitch), strongly coupled variation of the outer Euler coordinates can propagate through the innovation, corrected state, angular-rate state, and subsequent prediction. The controlled approximately stationary experiment confirms that this behaviour is associated with the coordinate representation rather than requiring large physical object motion.

The $SE(3)$ formulation instead represents rotational disagreement as a local Lie-algebra error derived from the relative orientation. Across six real-time trials and 21,109 frames, the complete tracking system achieved a mean rate of 29.35 FPS, while the $SE(3)$ filtering operations required only 0.973 ms per frame, corresponding to 2.86% of the total mean frame time. The principal contribution of the proposed method is therefore a geometrically consistent and representation-robust formulation of temporal 6D pose filtering that remains compatible with the FoundationPose++ pipeline and real-time RGB-D operation.